Student ID _____ Student Name _____

University of the South Pacific

Faculty of Science, Technology and Environment

School of Computing, Information and Mathematical Sciences

IS221 Web Applications Development

Mid Semester Test

Semester 1, 2023

Possible Marks:	**100**
Weight:	**20%**
Time Allowed:	**110** minutes
Number of Pages:	**11** (including this page)

Learning Outcomes:

2. Apply ethical practices to the development of the websites

Instructions:

1. Sections A, B, C: Answer ALL questions.
2. Write ALL the answers in the spaces provided in the Answer Booklet.
3. If you require extra space for your answers, use the last page.

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Section A: Matching Questions 20 Marks

Write the letter of the correct term that matches the given description in the space provided

A.	W3C	1.	___B_____ All display elements have been removed in favor of CSS.
B.	*HTML5*	**2.**	___A_____ Sets standards for HTML and other markup languages.
C.	*Universal selector*	**3.**	____L____Tell the browser which document version you are using .
D.	*Maintainability*	**4.**	_____C___ * {color: purple;}
E.	*Structure*	**5.**	____D____.Advantage of using CSS.
F.	*Type selector*	**6.**	___T_____ Information and user interface components must be perceivable by users.
G.	* *	**7.**	___H_____ Enables parts of text content to be treated differently from the surrounding text.
H.	*Inline*	**8.**	_____F___ H1 {color: red:}
I.	*Browser cache*	**9.**	____E____ Parts of the page such as the title, a heading, a block such as a division or paragraph, or a cell in a table.
J.	*Core Web fonts*	**10.**	___J_____ Set of fonts designed for specific use on the Web..
K.	*Definition lists*	11.	____I____ On return visits, your computer loads the files locally unless content has changed.
L.	*<?xml version = "1.0"?>*	12.	____G____ Break up a line at a certain point.
M.	*Div*	13.	_____Q___ Renders text in bold.
N.	*Hyperlink*	14.	___M_____Used to identify blocks which will have common style Properties.
O.	*Operability*	15.	_____K___ Use <dt> and <dd> instead of <li>
P.	*Domain name*	16.	____P____ An alias that points to your actual location on the Web server.
Q.	**	17.	___R_____ indicate a section of text that the author wants to identify as significant.
R.	**	18.	____S____ h1, h2 {color: red;}
S.	*Grouping selectors*	19.	___N_____ <a href="laptop.htm">laptop</a>
T.	*Perceivability*	20.	___O_____ User interface components must be operable by users.

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Section B: Coding Questions 40 Marks

1) The code below in Figure 1 represents the content a Two Trees Olive Oil website. using appropriate HTML5 semantic elements markup the code in Figure 1 below to make meaningful semantic HTML5 code. **[15 Marks]**

Two Trees Olive Oil

Home | About | Events | Shop | Blog | Contact

How We Make Our Olive Oil

Our extra-virgin olive oil (EVOO) is made with equal parts love and labor, and always with the utmost attention to quality, purity, and flavor. The process for making EVOO, the highest available grade of olive oil, is not easy. EVOO is produced entirely by mechanical means, unlike refined oils which are extracted through the use of chemicals or heat. The oil must pass a battery of chemical tests set by the California Olive Oil Council , as well as other organizations, and must pass taste tests by a panel of professional tasters who ensure the oil tastes like olives and has no taste flaws. In short, our olive oil goes through a lot before it arrives at the store in those pretty bottles.

Cultivating

Our olive trees are diligently pruned, watered, and monitored for pests while the fruit is growing. In addition, we apply organic farming techniques to ensure we can carry the “organic” label.

Harvesting

We harvest the greener, more unripe olives, to preserve quality and nutritional value. Harvesting is carried out by hand shaking the trees, or by using the newer mechanical harvesters, with the main goal of preventing damage to the fruit.

Cleaning

Within twelve hours of picking, harvested olives are transported to the mill where they are dropped into a hopper and move along a conveyor belt to be washed, sorted, and cleaned of branches and leaves.

Crushing

Inside the crusher, the olives are ground into a paste to facilitate the release of oil.

Frequently Asked Questions

How does extra virgin olive oil (EVOO) compare to regular olive oil?

EVOO is made from pure, cold-pressed olives and is certified as free of any defects in flavor or odor, when tested by a trained panel of experts. Regular olive oil is made from a blend of cold-pressed and processed oils.

Does olive oil get better over time?

Unlike wine, olive does not improve with age. For optimum health benefits and flavor, purchase EVOO within 12 to 18 months of harvest, and use within six months of opening.

How should I store olive oil?

To preserve flavor and integrity, store olive oil in a cool, dry, dark cupboard away from heat and light. Store between 57 to 70 degrees Fahrenheit.

What are polyphenols and how are they beneficial?

Polyphenols are compounds that function as antioxidants to protect cells against unwanted inflammation and disease. EVOO contains higher levels of polyphenols than other olive oils.

Testimonials

The best olive oil out there!
Jane, Massachusetts

Amazing flavor! Garlic infused is my favorite.
Sally, Arizona

Two Trees Olive Oil
22 Two Trees Lane
Santa Rosa, CA 95401
1-800-000-0000
info@twotreesoliveoil.com

Follow us: Facebook | Twitter

Figure 1: Two Trees Olive Oil Website

```
<body>
  <header>  1/2 MARK
    <h1>Two Trees Olive Oil</h1> 1/2 MARK
    <nav> 1/2 MARK
      <ul>
        <li><a href="#">Home</a></li>
        <li><a href="#">About</a></li>
        <li><a href="#">Events</a></li>      1/2 MARK
        <li><a href="#">Shop</a></li>
        <li><a href="#">Blog</a></li>
        <li><a href="#">Contact</a></li>
      </ul>
    </nav>
  </header>
  <article> 1/2 MARK
    <h2>How We Make Our Olive Oil</h2> 1/2 MARK

    <p>1/2MARK Our extra-virgin olive oil (<abbr title="Extra virgin olive
    oil">EVOO</abbr>)1/2MARK is made with equal parts love and labor, and always with the
    utmost attention to quality, purity, and flavor. The process for making <abbr title="Extra
    virgin olive oil">EVOO</abbr>, the highest available grade of olive oil, is not easy. <abbr
    title="Extra virgin olive oil">EVOO</abbr> is produced entirely by mechanical means,
    unlike refined oils which are extracted through the use of chemicals or heat. The oil must
    pass a battery of chemical tests set by the California Olive Oil Council (<abbr
    title="California Olive Oil Council">COOC</abbr>)1/2MARK, as well as other organizations,
    and must pass taste tests by a panel of professional tasters who ensure the oil tastes like
    olives and has no taste flaws. In short, our olive oil goes through a lot before it arrives at
    the store in those pretty bottles.</p>

    <h3>Cultivating</h3>1/2MARK

    <p>Our olive trees are diligently pruned, watered, and monitored for pests while the fruit
    is growing. In addition, we apply organic farming techniques to ensure we can carry the
    “organic” label.</p>1/2MARK

    <h3>Harvesting</h3>1/2MARK

    <p>We harvest the greener, more unripe olives, to preserve quality and nutritional value.
    Harvesting is carried out by hand shaking the trees, or by using the newer mechanical
    harvesters, with the main goal of preventing damage to the fruit.</p>1/2MARK

    <h3>Cleaning</h3>1/2MARK

    <p>Within twelve hours of picking, harvested olives are transported to the mill where they
    are dropped into a hopper and move along a conveyor belt to be washed, sorted, and
    cleaned of branches and leaves.</p> 1/2MARK
```

```
  <h3>Crushing</h3>1/2MARK
  <p>Inside the crusher, the olives are ground into a paste to facilitate the release of
  oil.</p>1/2MARK
</article>
<article> 1/2MARK
  <h2>Frequently Asked Questions</h2> 1/2MARK
    <dl>
      <dt>How does extra virgin olive oil (EVOO) compare to regular olive oil?</dt>
      <dd> EVOO is made from pure, cold-pressed olives and is certified as free of any
  defects in flavor or odor, when tested by a trained panel of experts. Regular olive oil is
  made from a blend of cold-pressed and processed oils.</dd>
      <dt>Does olive oil get better over time?</dt>
      <dd>Unlike wine, olive does not improve with age. For optimum health benefits and
  flavor, purchase EVOO within 12 to 18 months of harvest, and use within six months of
  opening.</dd>          1 MARK
      <dt>How should I store olive oil?</dt>
      <dd>To preserve flavor and integrity, store olive oil in a cool, dry, dark cupboard away
  from heat and light. Store between 57 to 70 degrees Fahrenheit.</dd>
      <dt>What are polyphenols and how are they beneficial? </dt>
      <dd>Polyphenols are compounds that function as antioxidants to protect cells against
  unwanted inflammation and disease. EVOO contains higher levels of polyphenols than
  other olive oils.</dd>
    </dl>
</article>
<aside> 1/2MARK
  <h2>Testimonials</h2> 1/2MARK
  <blockquote>
    <p>The best olive oil out there!</p>
    <footer><cite>Jane</cite>, Massachusetts</footer>
  </blockquote> 2 MARKS
  <blockquote>
    <p>Amazing flavor! Garlic infused is my favorite.</p>
    <footer><cite>Sally</cite>, Arizona</footer>
  </blockquote>
</aside>
<footer> 1/2 MARK
  <address> 1/2 MARK
    Two Trees Olive Oil<br>
    22 Two Trees Lane<br>
```

```
        Santa Rosa, CA 95401<br>
        <a href="tel:+118000000000">1-800-000-0000</a><br>
        <a href="mailto:info@twotreesoliveoil.com">info@twotreesoliveoil.com</a>
      </address>

      <p>Follow us: </p> 1/2 MARK
      <ul>
        <li><a href="ht tps://www.facebook.com/twotreesoliveoil/">Facebook</a></li>
        <li><a href="https://twitter.com/2treesoliveoil">Twitter</a></li>
      </ul>
    </footer>
  </body>
</html>
```

2) You have been employed to revamp the Exquisite Dining Experience website. Exquisite Dining Experience is a restaurant that prides itself in its authentic European, Asian and Island cuisines that are locally perfected. Use this scenario to answer the questions.:

a) Given the code for the recipe registration page depicted in pages 8 & 9, draw the web page when it is rendered. **[10 Marks]**

```
<%@ Page Title="Home Page" Language="VB" MasterPageFile="~/Site.Master"
AutoEventWireup="false"
    CodeFile="Subscribe.aspx.vb" Inherits="_Default" %>

<asp:Content ID="HeaderContent" runat="server" ContentPlaceHolderID="HeadContent">
    <style type="text/css">

.style1
{
            color: #0033CC;
text-align: center;
font-family: FangSong;
        }

.style2
{
            width: 600px;
        }

.style3
{
            height: 38px;
        }

.style5     {
        font-family: FangSong;
color: #0033CC;
    }
    </style>
</asp:Content>
<asp:Content ID="BodyContent" runat="server" ContentPlaceHolderID="MainContent">
    <h2 class="style1">
        Exquisite Dining Experiences Registration</h2>
    <p class="style1">
        <asp:Image ID="Image1" runat="server" ImageUrl="~/Images/images1.jpg"
            Width="100%" Height="200"/>
    </p>
    <p class="style1">
         </p>
    <table class="style2" align="center">
        <tr>
            <td>
                Name:</td>
            <td>
                <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
</td>
        </tr>
        <tr>
            <td>
                Email:</td>
            <td>
```

```
                    <asp:TextBox ID="txtEmail" runat="server"></asp:TextBox>
</td>
        </tr>
        <tr>
            <td>
                UserName:</td>
            <td>
                <asp:TextBox ID="txtUserName" runat="server"></asp:TextBox>
</td>
        </tr>
        <tr>
            <td>
                Password:</td>
            <td>
                <asp:TextBox ID="txtPassword" runat="server"
TextMode="Password"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>
                <h2 class="style5">Subscription Term:</h2></td>
            <td>
                <asp:DropDownList ID="ddlSubscriptionTerm" runat="server">
                    <asp:ListItem>Fortnightly</asp:ListItem>
                    <asp:ListItem>Monthly</asp:ListItem>
                    <asp:ListItem>Yearly</asp:ListItem>
                </asp:DropDownList>
            </td>
        </tr>
        <tr>
            <td>
                <h2 class="style5">Subscription For:</h2></td>
            <td>
                <asp:CheckBoxList ID="cblSubscriptions" runat="server"
RepeatColumns="0">
                    <asp:ListItem>Chef's Specials</asp:ListItem>
                    <asp:ListItem>Budget Recipes</asp:ListItem>
                    <asp:ListItem>French Cuisine Recipes</asp:ListItem>
                    <asp:ListItem>Italian Recipes</asp:ListItem>
                    <asp:ListItem>Mediteranian/Asian Fusion Cuisine</asp:ListItem>
                    <asp:ListItem>Deserts</asp:ListItem>
                    <asp:ListItem>Island Fusion Cuisine</asp:ListItem>
                    <asp:ListItem>Cocktails</asp:ListItem>
                </asp:CheckBoxList>
            </td>
        </tr>
        <tr>
            <td class="style3">
                </td>
            <td class="style3">
                <br />
                <asp:Button ID="btnSubscribe" runat="server" Text="Subscribe Me"
                    style="text-align: center" ForeColor="#0033CC" />
</td>
        </tr>
    </table>
    <p>
        <asp:Label ID="lblVerify" runat="server" ForeColor="#0033CC"></asp:Label>
    </p>

</asp:Content>
```

EXQUISITE DINING EXPERIENCES [Log In]

Home About

EXQUISITE DINING EXPERIENCES REGISTRATION

Name:

Email:

UserName:

Password:

SUBSCRIPTION PERIOD: Fortnightly

SUBSCRIPTION FOR:

- Chef's Specials
- Budget Cuisine
- French Cuisine
- Italian Cuisine
- Mediteranian/Asian Fusion
- Island Fusion
- Heavenly Deserts
- Cocktails

Subscribe Me

b) Analyse the use of CSS in the Style section on the code provided in Listing1 on pages 11, 12 and 13. Write a Style that will have all headings and paragraphs to be colored blue, centre aligned and font to be Calibri and assign background color to grey. **[5 marks]**

```
body {
        Color: blue;
        Align: centre;
        Font-family: Calibri;
        Background color: grey;
}
```

3) Given the source code below identify five specific errors, state the XHTML syntax rule that the error violates and state how to correct it using examples. **[10 Marks]**

```
<!DOCTYPE html
PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<HTML>
<head>
   <title>Table of Contents</title>
</head>
<body>

   <h1>
      <font face=sans-serif color="#993333" align=center>
         Introduction to HTML &minus; Table of Contents
      </font>
   </h1>
   <ol>
     <li>
      <dl>
         <dt><b>Introduction to the HTML</dt></b>
         <dd> Basic HTML Elements, CSS &alt; XHTML<br/></dd>
      </dl>
     <li>
      <dl><b>Body of an HTML Document</b>
      <li type=a>
         Organization of an HTML Document </dd><br />
        <dd> Lists</dd><br />
         <dd>   <ol type="square">
                         <li>DL (Glossary) Lists</li>
                  <li>Regular (UL, OL) Lists<br /></li>
            </ol>
         </dd>
          </dl>
   </li>
   <li>
      <a href="test">Bibliography and References</a>
      <br>
   </li>
   </ol>
   <hr width="85%" />
</body>
</HTML>
```

Error 1 _____

Syntax rule violated _____

Student ID _____ Student Name _____

Correction_____

Error 2_____

Syntax rule violated _____

Correction_____

Error 3_____

Syntax rule violated _____

Correction_____

Error 4_____

Syntax rule violated _____

Correction_____

Error 5_____

Syntax rule violated _____

Correction_____

Student ID _____ Student Name _____

Error 1

```
<!DOCTYPE html
PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
                    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-
                    transitional.dtd">
```

Error 2

<HTML>

```
<head>
   <title>Table of Contents</title>
</head>
<body>
```

Error 3

```
   <h1>
      <font face=sans-serif color="#993333" align=center>
Introduction to HTML &minus; Table of Contents
      </font>
   </h1>
   <ol>
     <li>
      <dl>
         <dt><b>Introduction to the HTML</dt></b>
         <dd> Basic HTML Elements, CSS &alt; XHTML<br/></dd>
      </dl>
     <li>
      <dl><b>Body of an HTML Document</b>
      <li type=a>
         Organization of an HTML Document </dd><br />
       <dd> Lists</dd><br />
         <dd>   <ol type="square">
                          <li>DL (Glossary) Lists</li>
                  <li>Regular (UL, OL) Lists<br /></li>
            </ol>
         </dd>
         </dl>
   </li>

   <li>
      <a href="test">Bibliography and References</a>
<br>
   </li>
   </ol>

   <hr width="85%" />

</body>
</HTML>
```

Error 4

Error 5

Error 1_____Missing document version in doctype declaration__**1MARK**_____

Syntax rule violated _____Document declaration includes document version, document type and namespace_**1/2MARK**_____

Correction___Add document version_____**1/2MARK** _____

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Error 2_____<HTML> - Element name in Capitals_**1 MARK**_____

Syntax rule violated ___All elements should be in lower case____**1/2MARK**_____

Correction_____<html>___**1/2MARK**_____

Error 3_____<font face=sans-serif color="#993333" align=center>_____**1 MARK**_
_____

Syntax rule violated _____All attribute values should be in quotation marks__**1/2MARK**__

Correction_____<font-face=”sans-serif align=”center”>__ **1/2MARK** _____

Error 4___<dt>Introduction to the HTML</dt></b ___**1 MARK**_____

Syntax rule violated _____All elements should be properly nested_____**1/2MARK** _

Correction___<dt>Introduction to the HTML</dt> _____**1/2MARK** _____

Error 5___as depicted ____**1 MARK** _____

Syntax rule violated _____Document must be well formed__**1/2MARK**_____

Correction_____Properly nest elements and use <dl> and <dd> tags instead of <li>____
1/2MARK

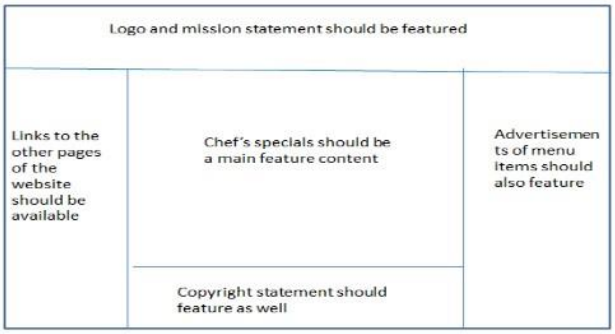
Section C: Short Answer Questions 40 Marks

1) You have been employed to re-design Exquisite Dinig Experience's website and are required to integrate the following site requirements:

- Logo and mission statement should be featured.
- Search box
- Links to the other pages of the website should be available
- Chef's specials should be a main feature content
- Advertisements of menu items should also feature
- Copyright statement should feature as well
- Links to all other pages should be available on all pages
- Uniform look and feel for all the pages, the only difference should be the actual conten

a) With the aid of wireframes, illustrate a potential template for the Home page.

[10 Marks]



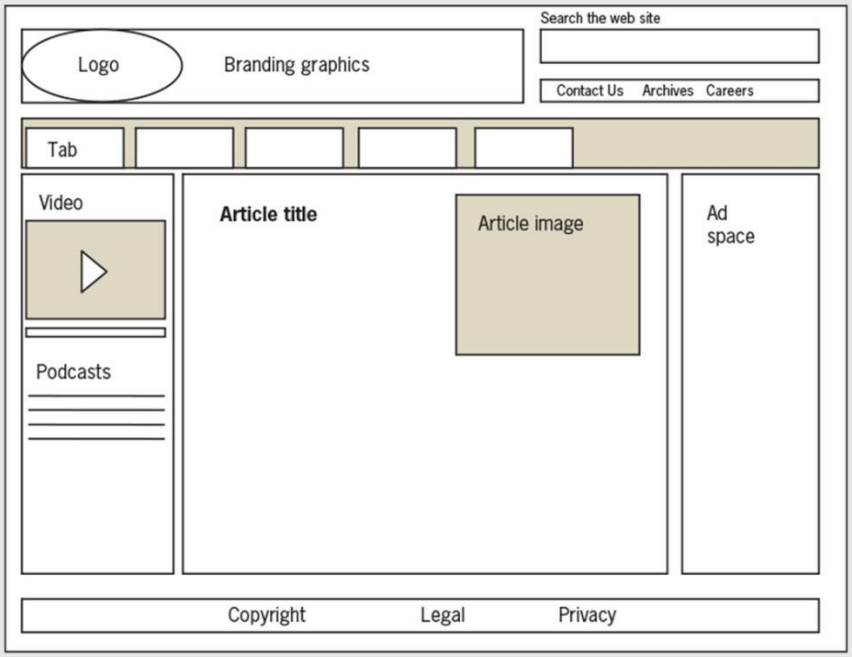
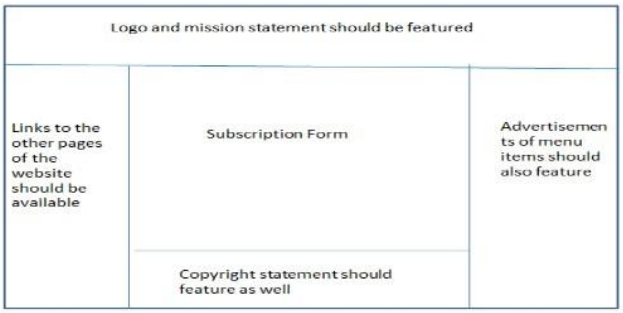


Figure 3-3: Sample wireframe for page layout
© 2015 Cengage Learning®

b) With the aid of wireframes illustrate how one of the content pages would look like.

[5 Marks]



c) Discuss how you would design for both Accessibility and Legibility **[8 Marks]**

Accessibility **4 MARKS**

Information and user interface should be perceivable to the users

User interface components should be operable by users

Information about the user interface its operations must be understandable by users Content shuld be robust enough to be interpreted by wide variety of agents.

Legibility **4 MARKS**

Choice of fonts should be easy to read and universally available for users' browsers.

Choice of colour should complement content

Use graphics sparingly for emphasis

Use of active whitespace for structure and visual enhancement of text

2) What is the difference between static and dynamic web page? **[5 Marks]**

Static page – contents remain same; pages that cannot change after the browser renders them**2.5 MARKS**

Dynamic page – data changes to reflect current data or a page that is customized to users preferences.(website interactes with DB**2.5 MARKS**

3) Effective navigation includes providing not only links to other pages in the Web site, but also cues to user's location. List the four navigation questions that users should be able to answer when navigating through your Web site. For each of the questions:

- Identify the type of information that the website is required to provide

[12marks]

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Question 1 *(1 Mark)*: **Where am I?**

Type of information: *(2Marks)*
The current page and the type of content they are viewing

_____
_____

Question 2 *(1 Mark)*: **Where can I go?**

Type of information: *(2 Marks)*
Where they are in relation to the rest of the Web site

_____
_____

Question 3 *(1 Mark)*: **How do I get there?**

Type of information: *(2 Marks)*
Consistent, easy-to-understand links

_____
_____

Question 4 *(1 Mark)*: **How do I get back to where I started?**

Type of information: *2 Marks)*
Alternatives to the browser's Back button that lets users return to their starting point

_____
_____